Statistical Learning-Classification

Project Title: APTOS 2019 Blindness Detection

Project Number: 1

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Your project falls into one of the following categories. Check the boxes which describe your project the best.

1. ☑ **Kaggle project.** Our project is a Kaggle competition.
   - This competition is active ☐ inactive ☑.
   - Our rank in the competition is 1329/2931.
   - The best Kaggle score in this competition is 0.936129, and our score is 0.894370.
2. ☐ **New algorithm.** We developed a new algorithm and demonstrated (theoretically and/or empirically) why our technique is better (or worse) than other algorithms.
3. ☐ **Application.** We applied known algorithm(s) to some domain.
   - ☐ We applied the algorithm(s) to our own research problem.
   - ☐ We tried to reproduce results of someone else's paper.
   - ☐ We used an existing implementation of the algorithm(s).
   - ☐ We implemented the algorithm(s) ourself.

Our most significant contributions are (List at most three):

(a) Demonstrated that an ensemble neural network of two EfficientNetB5's and a DenseNet121 with weighted soft voting achieved the highest score among individual state-of-the-art models, given the same data and preprocessing method

(b) Developed a preprocessing function that automatically crops and enhances the images to highlight clinically relevant features, and showed that our approach to cropping is not suitable for this problem

(c) Experimented on models that are trained with and without data augmentation, comprising of random flips and rotations, and concluded that data augmentation improved performance of all previously attempted models on the test dataset.

List the name of programming languages, tools, packages, and software that you have used in this project:

- Programming languages: Python 3.7, R
- Tools: Kaggle, Google Colab, Google Drive
- Packages: Keras, Tensorflow, NumPy, pandas, OpenCV, tqdm, Python Image Library (PIL), Matplotlib, Augmentor, json

1 Introduction

Diabetic retinopathy (DR) is a diabetic eye disease that damages the small blood vessel of the retina as a result of high blood glucose. It is the leading cause of blindness among working-aged adults. Since patients' vision do not deteriorate until the later stage of DR, early detection is difficult. The clinical approach to diagnose DR is through manual interpretation of retinal images.

Early detection can prevent or delay the onset of DR. At least 56% of new cases of DR could be reduced with proper and timely treatment, and monitoring of the eyes (Rohan T, 1989). However, conducting medical screening is difficult due to the following reasons:

1) Images taken at standard examinations are often noisy and poorly contrasted.
2) Current diagnosis relies on ophthalmologists to review images of the retina, which is inefficient. This problem is common in rural areas of developing countries where medical resources are lacking.
3) The complexity of DR diagnosis introduces grader variability, a common problem in human interpretation of imaging in the medical field. (Krause et al. 2018)

Image processing can reduce the above problems through image enhancement[1] and automatic analysis[2] of retina images. In 2019, APTOS (Asia Pacific Tele-Ophthalmology Society) and competition platform Kaggle created the APTOS 2019 Blindness Detection Competition and called for researchers to develop a five-class DR automatic diagnosing algorithm indicating severity of the disease.

2 Related Works

Convolutional neural networks (CNN) remain at the forefront in computer vision tasks since AlexNet's groundbreaking success in 2012 (Krizhevsky et al 2012). Recent contributions to the CNN architecture typically focus on sparsity and efficiency as their primary goals. In a popular 2016 paper by He et al, it was noted that the VGG neural networks (Liu et al 2015) are examples of deep neural networks that are notoriously difficult to train and easily overfit. As a result, the researchers proposed a novel mapping function in ResNet to improve optimization when learning. Such networks use *identity mapping* where the input of subsequent layers is the activated sum of the input and output of the previous layers. Mathematically, when x is the input and F_θ is the functional representation of a block with weights θ, the output is $\sigma(F_\theta(x) + x)$. With the presence of an identity mapping, the error signals can be more easily propagated through the layers since having identity mappings lowers nonlinearity across layers. DenseNet further extends this concept by adding $\frac{L(L+1)}{2}$ identity mapping connections between layers and reducing the number of filters (Huang et al. 2017) where L is the number of layers.

ResNet and DenseNet are examples of depth scaling where it can be extended to 152 and 264 layers respectively, as were used in the ImageNet competition (He et al 2016). EfficientNet, introduced in a paper by Tan and Le, proposed scaling in multiple dimensions (depth, width and resolution) called *compound scaling*. The scaling parameter ϕ dictates the growth rate of depth $d = \alpha^\phi$, width $w = \beta^\phi$ and resolution $r = \gamma^\phi$, such that under certain constraints, the floating point operations (FLOPS) required will experience a growth rate of 2^ϕ. EfficientNet uses *mobile inverted bottleneck* layer as its main building block (Tan and Le 2019) but varies the dimensions and depth of the layers according to the predetermined ϕ. Empirical results suggest that EfficientNet is among the top performers for the ImageNet dataset as of 2020 but with significantly lower parameters and FLOPS.

3 Exploratory Data Analysis

The dataset consists of high-resolution retina images with various shapes under a variety of imaging conditions. There are 3662 training images and 1928 test images. Having a small dataset supports the proposition

[1] Image processing can increase contrast and reduce noise in the images, which aids manual diagnosis. (Walter et al 2002)

[2] Image processing automates analysis of DR, which includes detection of pathologies like microaneurysms, and detection of features in the retina such as vascular trees. It reduces examination time and reliance on specialists. This facilitates mass screening in places where specialists are lacking. (Walter et al 2002)

that data augmentation and transfer learning would be beneficial. Images are rated by ophthalmologist on a scale of 0 to 4: 0 - No DR, 1 - Mild, 2 - Moderate, 3 - Severe, 4 - Proliferative DR

From Figure 1, the dataset is highly imbalanced: half of the training samples are in class “no DR”. Thus, oversampling or undersampling is required to balance the distribution of classes. In addition, class “moderate DR” is a lot higher than class “Mild”, “Severe” and “Proliferative DR”.

Figure 2 shows two random images from each class. The image brightness, colour concentration and the height and width ratio of images has significant variation. For example, the second sample in class “no DR” is more zoomed in and dimmer than the first sample in the same class. In addition, image size is too large which necessitates the use of autocropping, padding and adjustment to image brightness.

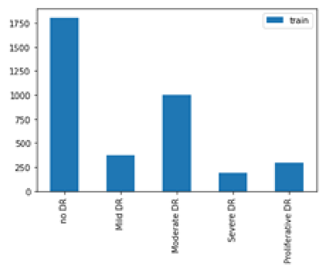


Figure 1: Distribution of training sample

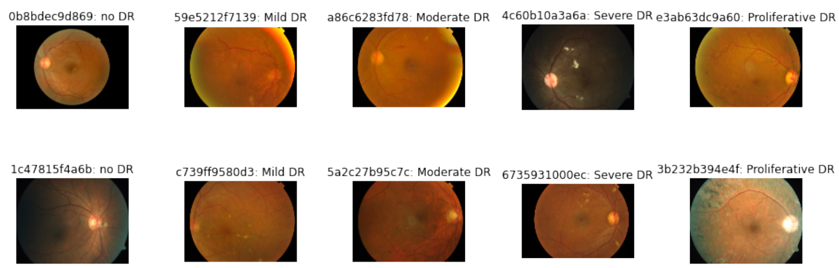


Figure 2: Sample images of training data

Figure 3 displays the differences of image dimension between train and test set. It shows that the distribution of image dimension between train and test data vastly differs, which might contribute to a significant difference between the cross validation score and test score.

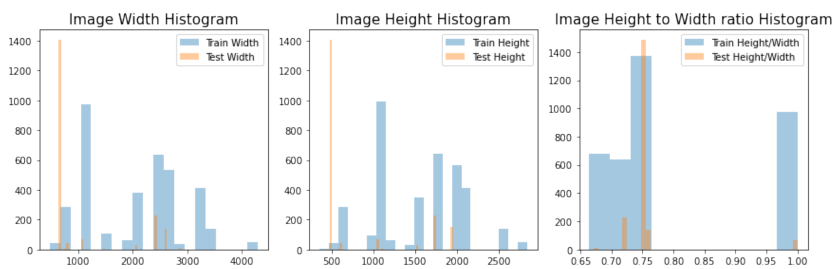


Figure 3: Distribution of height, width, height-to-width ratio in train and test set

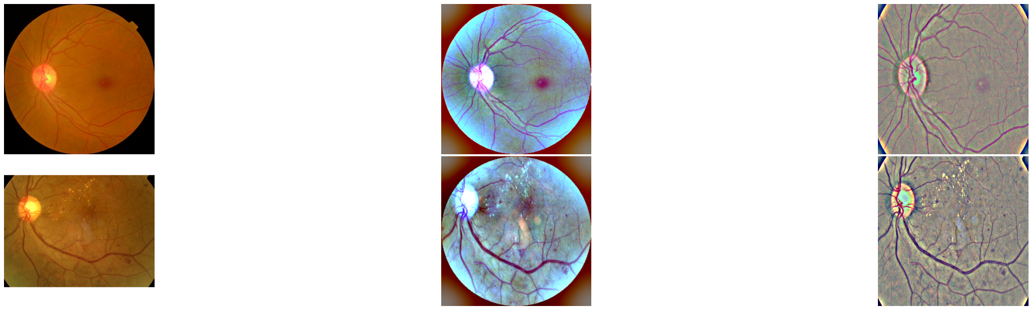
4 Methodology

4.1 Image Preprocessing

The data provided are of different sizes, aspect ratios and degrees of cropped edges. To ensure consistency in the images, we developed an algorithm to dynamically crop and scale all images to a predefined format. Images are cropped to an aspect ratio of 4:3 and deliberately zoomed such that only 1% of the background remains. The images are scaled to a dimension of 256×256. The final step is to standardize contrast of the images and improve feature prominence with linear blending[3]. The remaining details of the algorithm are described as pseudocode in Algorithm 1. The constant c_{ratio} is empirically determined to be 1.5 based on various test images.

We compare our preprocessed sample images to the original image and the output from the preprocessing function by the winner of APTOS 2015 (Graham 2015). The following two examples in Figure 4 illustrate the typical outputs from the two preprocessing functions.

[3]The linear blending section is inspired by the winner of APTOS 2015 (Graham 2015)



(a) Original images (b) Images with Graham's preprocessing (c) Images with our preprocessing

Figure 4: Example of preprocessing on sample images (top row is train data and bottom row is test data)

4.2 Experiment Setup

The experiment is set up in two phases: determining the best preprocessing algorithm and determining the best model. In the first phase, we fix the choice of neural network architecture to be EfficientNetB5. This choice is based on reported findings from the EfficientNet paper which reaches state-of-the-art accuracy on ImageNet (Tan and Le 2019). Two models are trained with images produced by our preprocessing functions and Graham's preprocessing function (winner's preprocessing function) respectively. The two models have the same hyperparameter configuration. Upon submission of both models for evaluation by Kaggle, the model that was trained with our preprocessing function received a score of 0.606565 while the model that was trained with Graham's preprocessing function received a score of 0.835497. Since the difference in performance is significant, it can be deduced that Graham's preprocessing function is superior and will be used for subsequent experiments for determining the best model.

For the second phase, several state-of-the-art architectures for computer vision tasks are nominated: ResNet, DenseNet and EfficientNet. Instead of training models from scratch, we adopt transfer learning as a method to improve performance and increase the rate of convergence. The pretrained models, which were trained on the ImageNet dataset[4], are obtained from Keras. Historically, ensemble models have excelled in Kaggle competitions due to their incredible predictive performance in practical scenarios (Sun and Pfahringer 2011). Therefore, we also conduct trials on ensemble neural network models with weighted soft voting. Additionally, data augmentation is applied on some trials to improve the robustness of the model as 85% of the test data are unseen. The parameters for data augmentation are: random horizontal and vertical flipping with a probability of 0.5 and uniformly distributed rotation from $0°$ to $360°$.

5 Kaggle Ranking Results

The models we experimented were submitted to Kaggle: APTOS 2019 Blindness Detection competition for evaluation. The models are evaluated using 85% of the test data, which is used to determine the rankings on the private leaderboard. Private leaderboard reflects the final standings. The evaluation metric is quadratic weighted kappa[5]. Below is a subset of the models we experimented with corresponding performances:

6 Discussion

After finishing all the experiments, we reviewed our models and summarized several improvements that could be potentially made to increase model performance:

- **External Data** The training data only contains 3662 images and is severely imbalanced with approximately 50% of the data labelled as class 0. To get a more robust model, external data could be used

[4]An image database that is typically used to benchmark computer vision performance.

[5]The quadratic weighted kappa is calculated between the scores assigned by the human rater and the predicted scores. It measures the agreement between two ratings and typically varies from 0 (no agreement) to 1 (complete agreement).

Model	Private Score	Ranking (out of 2931 teams)
ResNet50	0.792417	2376
DenseNet121	0.840077	2217
DenseNet201	0.802247	2344
EfficientNetB3	0.672945	2484
EfficientNetB5	0.835497	2246
EfficientNetB5 + own preprocessing method	0.606565	2516
DenseNet121 + data augmentation	0.844387	2188
EfficientNetB5 + data augmentation	0.872725	1922
Ensemble of 2 × EfficientNetB5 + data augmentation	0.892283	1395
Ensemble of 2 × EfficientNetB5 and DenseNet121 + data augmentation	0.894370	1329

Table 1: Kaggle Score and Ranking

to train the models. Images from previous competitions such as APTOS 2015 could be used (given that external data is allowed in this Kaggle competition).

- **Synthetic Sampling** To avoid undersampling majority classes, synthetic sampling can be used in conjunction with our data augmentation techniques to generate additional images for minority classes. This approach is essentially oversampling but does not have the caveat of overfitting on minority classes since the generated images are different from the original images. Additionally, we can extend upon our existing data augmentation techniques by including zooming, width and height shifting and shearing since experimental results suggest that data augmentation improves results.
- **Ensemble Method** This technique combines multiple models to produce improved results. It helps reduce bias and variance and thus improving the performance of the model. Our approach is to apply weighted soft voting on the models with the best performance. More advanced ensemble methods such as bagging and boosting could be implemented to improve the model, although more computational power is required.
- **Image Size** We found that different image sizes can lead to significant changes in validation score and test score for different CNN architectures. During our experiments, EfficientNetB5 performed best under size of 456 × 456 whereas DenseNet121 performed best under size of 224 × 224. Different combinations of CNN architectures and image sizes could be experimented to improve the model.

7 Conclusion

The extensive experimentation on different model architectures, preprocessing approaches, data augmentation and ensemble combinations yielded various insights on the nature of diabetic retinopathy diagnosis. A clear observation is the improvement in results by augmenting data being fed into the models. Another observation was the significant improvement in score when incorporating ensemble methods (jumping an average of 560 rankings from our best model without ensemble). Nevertheless, this series of experimentations cannot decisively conclude the best hyperparameters for certain important factors such as image size. While our best model suggests good agreement with human raters, it is still limited as a black-box model that fails to provide insights and justification on its predictions. In the relentless pursuit of superior predictive power, we should not lose sight that the healthcare industry deems interpretability of diagnosis as an imperative element in such situations.

8 References

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Appendix

Algorithm 1 Preprocessing Algorithm

Input 1: variable dimension image (matrix) I with RGB channels
Input 2: c_{ratio}, ratio to crop away from the top of the image

$\ell_h, \ell_v \leftarrow$ horizontal and vertical limits of the non-background regions respectively
$I' \leftarrow$ crop I at ℓ_h and ℓ_v and discard regions outside the bounding box
$v \leftarrow$ smallest index of non-dark pixel on the leftmost column of the image I'
$r \leftarrow$ compute radius of circle with the Euclidean distance of coordinate $(v, 0)$ to the midpoint
$I'' \leftarrow$ crop the top and bottom of the image I' by $r - \sqrt{r^2 - \left(\frac{c_{\text{ratio}} \cdot r}{2}\right)^2}$
$I''' \leftarrow$ crop the left and right of the image I'' so that the aspect ratio is $4:3$
$O \leftarrow$ resize I''' to a dimension of 256×256
$O \leftarrow$ apply linear blending with O and a negative Gaussian blur filter ($O \leftarrow O - \text{blur}(O) + 128$)

return O